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# 1. Requirements Table

| Req ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| FR-001 | Functional | The system shall generate collection routes for garbage trucks. | High | Route is displayed to the driver. | Main function |
| FR-002 | Functional | The system shall receive fill-level data from smart bins. | High | Bin status is updated. | Keeps data current |
| FR-003 | Functional | The system shall allow the supervisor to view routes. | Medium | Route opens successfully. | Easy monitoring |
| FR-004 | Functional | The system shall allow citizens to submit complaints. | Medium | Complaint is saved. | Improves service |
| FR-005 | Functional | The system shall notify drivers about new routes. | Medium | Driver receives notification. | Better communication |
| NFR-001 | Non-Functional | The system shall generate routes within 45 seconds. | High | Route generated in under 45 seconds. | Fast performance |
| NFR-002 | Non-Functional | The system shall be available 99% of the time. | Medium | System uptime is at least 99%. | Reliable service |

# 2. Use Case Flow

### Use Case Flow Specification

| Field | Details |
|---|---|
| **Use Case Name** | Generate Route |
| **Primary Actor** | Supervisor |
| **Goal** | Generate a garbage collection route for truck drivers. |
| **Preconditions** | 1. Supervisor is logged into the system. |
| **Postconditions** | 1. Route is generated.<br>2. Truck driver is notified. |

### Main Flow

| Step | Action |
|---|---|
| 1 | Supervisor logs into the system. |
| 2 | Supervisor selects **Generate Route**. |
| 3 | The system generates the collection route. |
| 4 | The supervisor views the generated route. |
| 5 | The system notifies the truck driver. |
| 6 | The truck driver views the route. |
| 7 | Garbage collection begins. |

### Alternate Flow

| Step | Action |
|---|---|
| 3A | Route generation fails. |
| 3A.1 | The system displays an error message. |
| 3A.2 | The supervisor tries again. |
| 3A.3 | The system generates the route successfully. |

# 3. Use Case Diagram from StarUml:

